



TechBiz Academy

Week 3 · Task 9

Enumeration _ Digging Deeper into Services

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Tools Used:

- Nmap
- enum4linux
- smbclient
- Netcat (nc)

Objective

The objective of this lab was to perform service enumeration on the SMB service running on the target machine. The task involved identifying available SMB shares, enumerating users, obtaining operating system information, and grabbing service banners from FTP and SSH.

Lab Environment

Component	Details
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2
Target IP	192.168.1.6
Network Type	VirtualBox Internal Network

Tools Used

- Nmap
 - enum4linux
 - smbclient
 - Netcat (nc)
-

Task Steps

Step 1 – Port Scanning

First, an Nmap scan was performed to identify open ports and running services on the target.

Command

```
nmap -Pn 192.168.1.6
```

Result

The scan revealed multiple open ports including:

- FTP (21)
- SSH (22)
- Telnet (23)
- SMTP (25)
- DNS (53)
- HTTP (80)
- SMB (139,445)
- MySQL (3306)
- PostgreSQL (5432)
- VNC (5900)

The SMB service was selected for further enumeration.

Figure 1: Nmap Scan Output



```
PORT      STATE SERVICE      VERSION
445/tcp    open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 08:00:27:09:CA:FC (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.13 seconds

(hamza@hamza)-[~]
$ nmap -Pn 192.168.1.6
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-21 02:53 -0400
Nmap scan report for 192.168.1.6
Host is up (0.00051s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:09:CA:FC (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.20 seconds

(hamza@hamza)-[~]
```

Step 2 – SMB Anonymous Enumeration

Anonymous SMB login was tested using smbclient.

Command

```
smbclient -L //192.168.1.6 -N
```

Result

Anonymous login was successful.

The following shares were discovered:

- print\$
- tmp
- opt
- IPC\$
- ADMIN\$

Figure 2: SMB Share Enumeration

A screenshot of a Linux terminal window titled 'hamza@hamza: ~/Arzens_Test'. The terminal shows the execution of 'smbclient -L //192.168.1.6 -N', which successfully lists several SMB shares: print\$ (Printer Drivers), tmp (oh noes!), opt, IPC\$ (IPC Service), and ADMIN\$ (IPC Service). Following this, 'enum4linux 192.168.1.6' is run, displaying target information for 192.168.1.6, including RID ranges (500-550, 1000-1050), username, password, and known usernames (administrator, guest, krbtgt, domain admins, root, bin, none). The terminal also shows the domain/workgroup name as WORKGROUP.

```
(hamza@hamza)-[~/Arzens_Test]
$ smbclient -L //192.168.1.6 -N
Anonymous login successful

Sharename      Type      Comment
-----
print$         Disk      Printer Drivers
tmp            Disk      oh noes!
opt            Disk
IPC$           IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
ADMIN$         IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))

Reconnecting with SMB1 for workgroup listing.
Anonymous login successful

Server          Comment
-----
the default we... the default we...
Workgroup       Master
WORKGROUP      METASPLOITABLE

(hamza@hamza)-[~/Arzens_Test]
$ enum4linux 192.168.1.6
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Sun Jul 19 04:15:54 2026

===== ( Target Information ) =====
Target ..... 192.168.1.6
RID Range ..... 500-550,1000-1050
Username ..... 
Password ..... 
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.1.6 ) =====

[+] Got domain/workgroup name: WORKGROUP

Looking up status of 192.168.1.6
METASPLOITABLE <00> =
```

Step 3 – SMB Enumeration using enum4linux

The enum4linux tool was executed to enumerate detailed SMB information.

Command

```
enum4linux 192.168.1.6
```

Figure 3: enum4linux – Target Information



```
hamza@hamza: ~/Arzens_Test
$ enum4linux 192.168.1.6
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Sun Jul 19 04:15:54 2026

===== ( Target Information ) =====
Target ..... 192.168.1.6
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.1.6 ) =====
[+] Got domain/workgroup name: WORKGROUP file (located at /var/www/html/index.html) before continuing to operate
your HTTP server.

===== ( Nbtstat Information for 192.168.1.6 ) =====
Looking up status of 192.168.1.6
METASPLOITABLE <00> - B <ACTIVE> Workstation Service
METASPLOITABLE <03> - B <ACTIVE> Messenger Service
METASPLOITABLE <20> - B <ACTIVE> File Server Service
.._MSBROWSE_.. <01> - <GROUP> B <ACTIVE> Master Browser
WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
WORKGROUP <1d> - <GROUP> B <ACTIVE> Master Browser
WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections

MAC Address = 00-00-00-00-00-00

===== ( Session Check on 192.168.1.6 ) =====
[+] Server 192.168.1.6 allows sessions using username '', password ''

===== ( Getting domain SID for 192.168.1.6 ) =====
Domain Name: WORKGROUP
Domain Sid: (NULL SID)
```

Information Discovered

- Host Name: **METASPLOITABLE**
- Workgroup: **WORKGROUP**
- Null session allowed
- SMB service detected

Figure 4: Operating System and User Enumeration



```
[*] Can't determine if host is part of domain or part of a workgroup

***** ( OS information on 192.168.1.6 )*****

[*] Can't get OS info with smbclient

[*] Got OS info for 192.168.1.6 from srmvinfo:
METASPLOITABLE Mk Sv PrQ Unix NT SNT metasploitable server (Samba 3.0.20-Debian)
platform_id      : 500
OS version       : 4.9
server type      : 0x9a03

***** ( Users on 192.168.1.6 )*****

Index: 0x1 RID: 0x1f3 acb: 0x00000011 Account: games Name: games Desc: (null)
Index: 0x2 RID: 0x1f5 acb: 0x00000011 Account: nobody Name: nobody Desc: (null)
Index: 0x3 RID: 0x4ba acb: 0x00000011 Account: bind Name: (null) Desc: (null)
Index: 0x4 RID: 0x402 acb: 0x00000011 Account: proxy Name: proxy Desc: (null)
Index: 0x5 RID: 0x4b4 acb: 0x00000011 Account: syslog Name: (null) Desc: (null)
Index: 0x6 RID: 0x4ba acb: 0x00000010 Account: user Name: Just a user, lol, Desc: (null)
Index: 0x7 RID: 0x42a acb: 0x00000011 Account: www-data Name: www-data Desc: (null)
Index: 0x8 RID: 0x3e8 acb: 0x00000011 Account: root Name: root Desc: (null)
Index: 0x9 RID: 0x2fa acb: 0x00000011 Account: news Name: news Desc: (null)
Index: 0xa RID: 0x4c8 acb: 0x00000011 Account: postgres Name: PostgreSQL administrator, Desc: (null)
Index: 0xb RID: 0x3ec acb: 0x00000011 Account: bin Name: bin Desc: (null)
Index: 0xc RID: 0x3f8 acb: 0x00000011 Account: mail Name: mail Desc: (null)
Index: 0xd RID: 0x4c8 acb: 0x00000011 Account: distccd Name: (null) Desc: (null)
Index: 0xe RID: 0x4ca acb: 0x00000011 Account: proftpd Name: (null) Desc: (null)
Index: 0xf RID: 0x4b2 acb: 0x00000011 Account: dhcp Name: (null) Desc: (null)
Index: 0x10 RID: 0x3ea acb: 0x00000011 Account: daemon Name: daemon Desc: (null)
Index: 0x11 RID: 0x4ba acb: 0x00000011 Account: sshd Name: (null) Desc: (null)
Index: 0x12 RID: 0x3f4 acb: 0x00000011 Account: man Name: man Desc: (null)
Index: 0x13 RID: 0x3f6 acb: 0x00000011 Account: lp Name: lp Desc: (null)
Index: 0x14 RID: 0x4c2 acb: 0x00000011 Account: mysql Name: MySQL Server, Desc: (null)
Index: 0x15 RID: 0x43a acb: 0x00000011 Account: gnats Name: Gnats Bug-reporting System (admin) Desc: (null)
Index: 0x16 RID: 0x4b8 acb: 0x00000011 Account: libuid Name: (null) Desc: (null)
Index: 0x17 RID: 0x42c acb: 0x00000011 Account: backup Name: backup Desc: (null)
Index: 0x18 RID: 0x4b8 acb: 0x00000010 Account: msfadmin Name: msfadmin, Desc: (null)
Index: 0x19 RID: 0x4c8 acb: 0x00000011 Account: telnetd Name: (null) Desc: (null)
Index: 0x1a RID: 0x3ee acb: 0x00000011 Account: sys Name: sys Desc: (null)
```

OS Information

- Samba Server Version:
Samba 3.0.20-Debian
- Platform ID:
500
- Operating System Version:
4.9

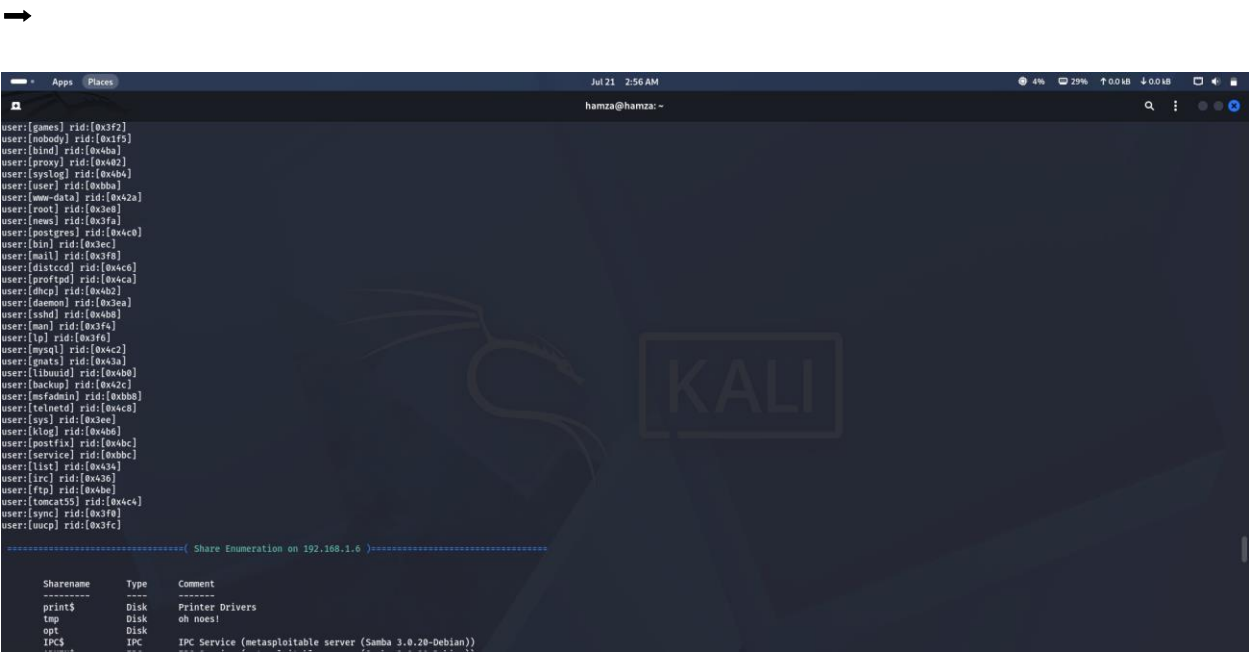
Enumerated Users

Several user accounts were identified, including:

- root
- postgres
- mysql
- ftp
- sshd
- daemon
- www-data
- msfadmin
- backup
- telnetd
- syslog
- news
- nobody

This confirms that user enumeration through SMB was successful.

Figure 5: SMB Share Enumeration



Shares Found

Share	Description
print\$	Printer Drivers

Share	Description
tmp	Temporary Share
opt	Optional Files
IPC\$	IPC Service
ADMIN\$	Administrative Share

Anonymous access was allowed on the **tmp** share.

Figure 6: Password Policy Information



```

-----
print$      Disk      Printer Drivers
tmp         Disk      oh noes!
opt         Disk
IPC$        IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
ADMIN$      IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
Reconnecting with SMB1 for workgroup listing.

Server      Comment
-----
Workgroup    Master
WORKGROUP   METASPLOITABLE

[*] Attempting to map shares on 192.168.1.6
//192.168.1.6/print$ Mapping: DENIED Listing: N/A Writing: N/A
//192.168.1.6/tmp Mapping: OK Listing: OK Writing: N/A
//192.168.1.6/opt Mapping: DENIED Listing: N/A Writing: N/A
[E] Can't understand response:
NT_STATUS_NETWORK_ACCESS_DENIED listing \*
//192.168.1.6/IPC$ Mapping: N/A Listing: N/A Writing: N/A
//192.168.1.6/ADMIN$ Mapping: DENIED Listing: N/A Writing: N/A

*****[ Password Policy Information for 192.168.1.6 ]*****
Password:

[*] Attaching to 192.168.1.6 using a NULL share
[*] Trying protocol 139/SMB...
[*] Found domain(s):
    [*] METASPLOITABLE
    [*] Builtin
[*] Password Info for Domain: METASPLOITABLE
    [*] Minimum password length: 5
    [*] Password History length: None

```

Password Policy Findings

- Minimum Password Length: 5
- Password History Length: **None**
- Password Complexity: Not enforced

These settings indicate weak password policies.

Step 4 – FTP Banner Grabbing

Banner grabbing was performed using Netcat.

Command

```
nc 192.168.1.6 21
```

Result

Banner received:

```
220 (vsFTPD 2.3.4)
```

Step 5 – SSH Banner Grabbing

Command

```
nc 192.168.1.6 22
```

Result

Banner received:

```
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
```

Figure 7: FTP and SSH Banner Grabbing

A screenshot of a Linux terminal window. The window title bar shows 'Apps', 'Places', 'Jul 21 3:01 AM', and system status icons (8% battery, 29% network, 0.0 kB upload/download). The terminal prompt is 'hamza@hamza: ~'. The first command executed is 'nc 192.168.1.6 21', which returns the banner '220 (vsFTPD 2.3.4)'. The second command is 'nc 192.168.1.6 22', which returns the banner 'SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1'. Both results are preceded by the prompt '(hamza@hamza)-[~]' and the command '\$'.

```
enum4linux complete on Tue Jul 21 02:55:12 2026

(hamza@hamza)-[~]
$ nc 192.168.1.6 21
220 (vsFTPD 2.3.4)
^C

(hamza@hamza)-[~]
$ nc 192.168.1.6 22
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
^C
```

Summary of Findings

Category	Result
Target IP	192.168.1.6
Hostname	METASPLOITABLE
Workgroup	WORKGROUP
SMB Version	Samba 3.0.20-Debian
FTP Version	vsFTPd 2.3.4
SSH Version	OpenSSH_4.7p1 Debian
Anonymous Login	Successful
Users Enumerated	Yes
SMB Shares Found	print\$, tmp, opt, IPC\$, ADMIN\$

Security Observations

- Anonymous SMB access is enabled.
 - Multiple user accounts can be enumerated.
 - The Samba version is outdated.
 - Weak password policy is configured.
 - Service versions are exposed through banner grabbing.
 - The system exposes several unnecessary network services, increasing the attack surface.
-

Learning Outcomes

After completing this task, I learned how to:

- Perform service enumeration using Nmap.
 - Enumerate SMB shares using smbclient.
 - Use enum4linux to gather detailed SMB information.
 - Identify hostnames, workgroups, users, and shares.
 - Perform banner grabbing with Netcat.
 - Analyze service versions and identify potential security weaknesses.
-

Learning Resources

During this lab, the following educational resources were used to understand the concepts of network scanning, SMB enumeration, and service discovery.

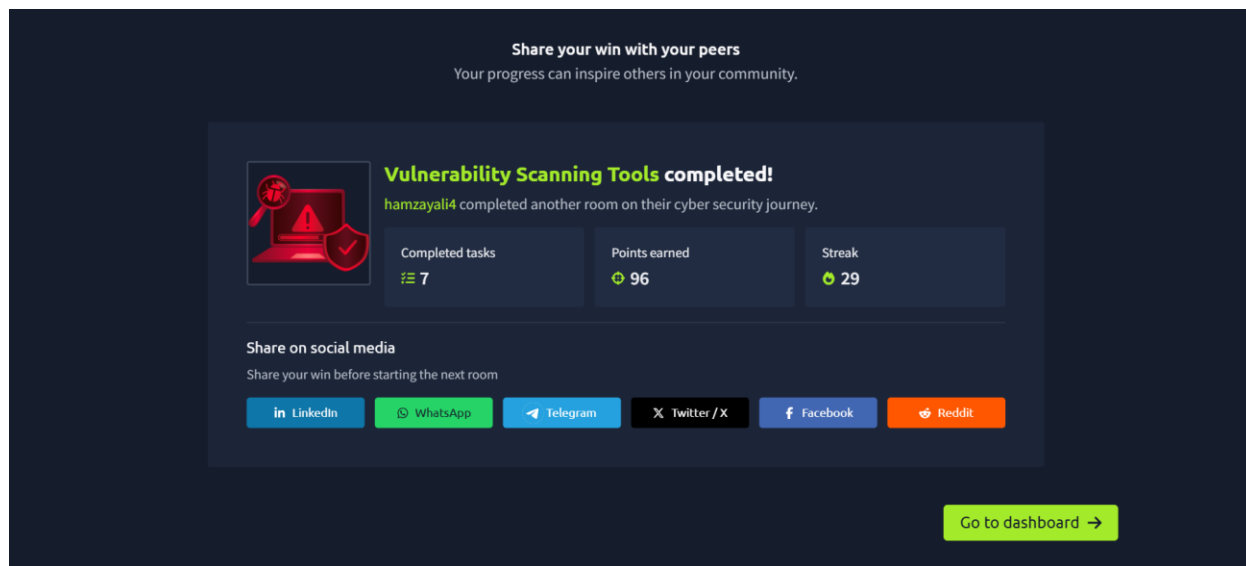
These resources helped in learning the practical usage of Nmap, enum4linux, SMB enumeration techniques, and service banner grabbing.

Resource 1 – TryHackMe: Vulnerability scanning tools

Purpose

This room provides hands-on practice for scanning and enumeration using industry-standard tools such as Nmap. It explains how to identify open ports, services, and vulnerabilities before moving to further penetration testing.

Screenshot

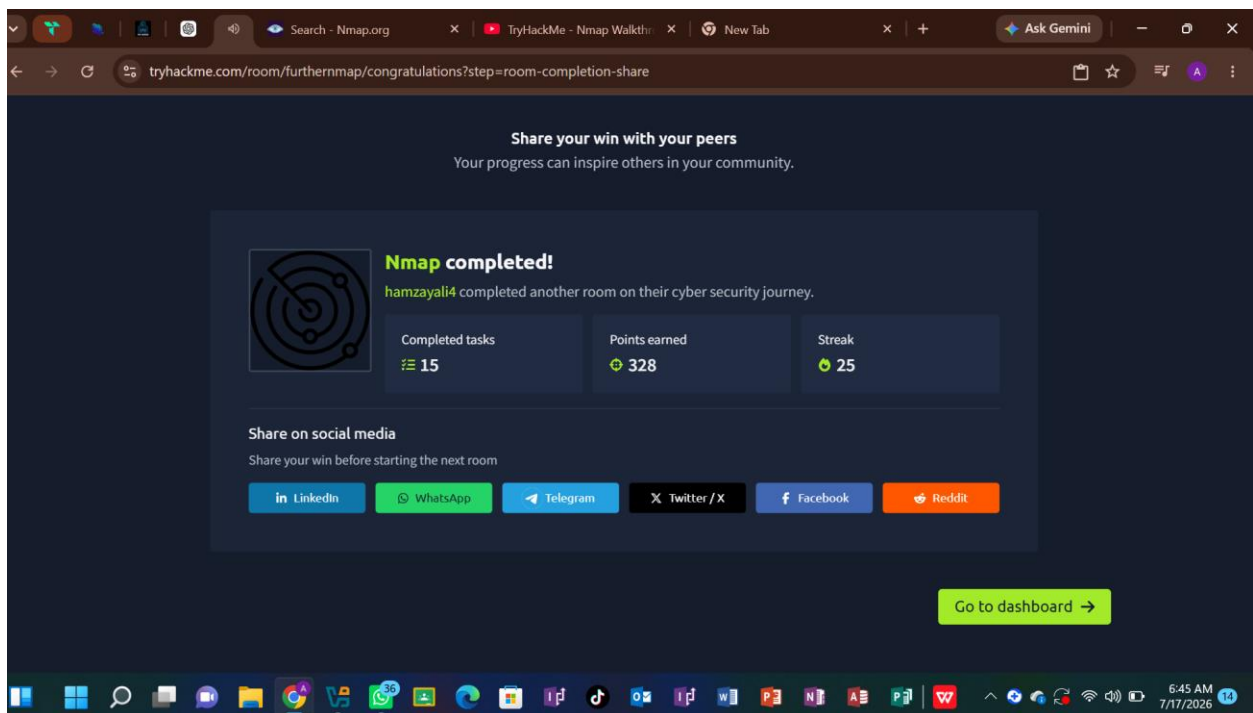


Resource 2 – TryHackMe: Nmap

Purpose

The TryHackMe Nmap room explains various scanning techniques, service detection, version detection, operating system identification, and scripting with NSE. It helped understand how to perform accurate network reconnaissance.

Screenshot

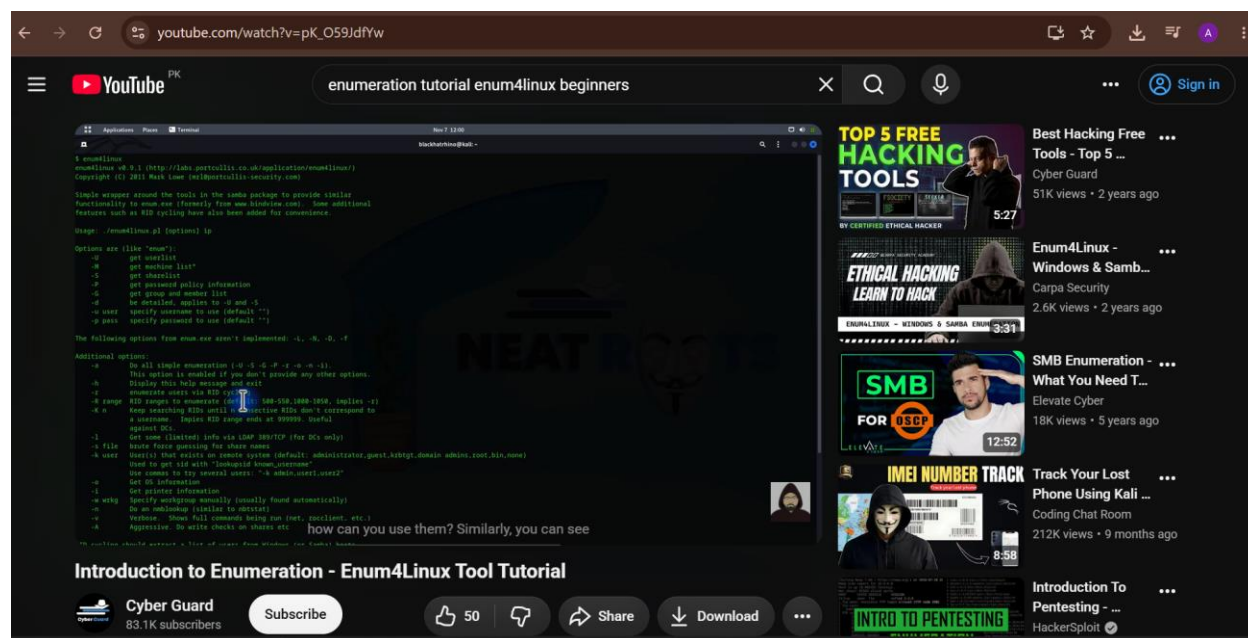


Resource 3 – YouTube: Enumeration Tutorial (enum4linux for Beginners)

Purpose

This tutorial demonstrated how to use **enum4linux** for SMB enumeration, including discovering workgroups, usernames, shared folders, password policies, and operating system information.

Screenshot

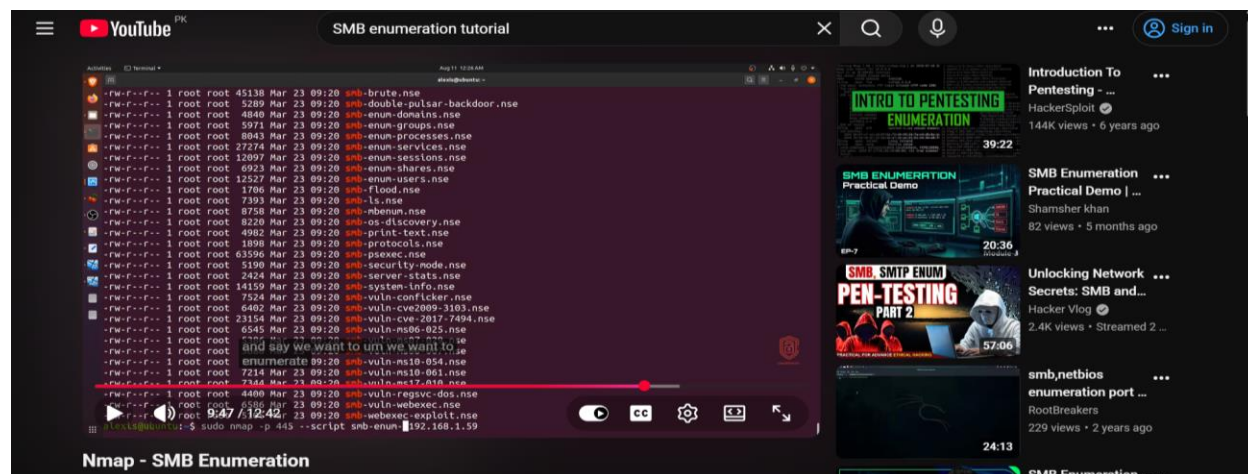


Resource 4 – YouTube: SMB Enumeration Tutorial

Purpose

This tutorial explained SMB enumeration techniques, anonymous share access, SMB version identification, share enumeration, and user enumeration using practical examples.

Screenshot



References

1. TryHackMe. *Vulniversity Room*. <https://tryhackme.com/room/vulniversity>
2. TryHackMe. *Nmap Room*. <https://tryhackme.com/room/furthernmap>
3. Nmap Project. *Official Documentation*. <https://nmap.org>
4. enum4linux Documentation. <http://labs.portcullis.co.uk/application/enum4linux/>
5. Samba Documentation. <https://www.samba.org/>
6. YouTube. *Enumeration Tutorial – enum4linux for Beginners*
7. YouTube. *SMB Enumeration Tutorial*

